



Fundamentals of Business Finance

Tutorial A

Activity 1



Please open this week worksheet and formula sheet

Open a web browser and go to nearpod.com. Login with the code displayed on our screen

Share what you know about the concepts

Activity 2



What are the common characteristics between an annuity and perpetuity?

What are the different characteristics between an annuity and perpetuity?

Activity 2



What are the common characteristics between an annuity and perpetuity?

- a series of equal payments
- made at equal intervals
- with the first payment at the end of the first period

What are the different characteristics between an annuity and perpetuity?

- Annuity: Will end one day -> The FV is occurring at the same time point as the last PMT of the Cash flow
- Perpetuity: Last forever -> No FV

Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ? Would Andrew be able to buy his first mobile phone?

Step 1: Identify the known variables and unknown variables

$I =$

$N =$

$PMT =$

$PV =$

$FV =$

Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ? Would Andrew be able to buy his first mobile phone?

Step 1: Identify the known variables and unknown variables

$$I = 5\%/12$$

$$N = 2 \times 12$$

$$PMT = -50$$

$$PV = 0$$

$$FV = ?$$

Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ?
Would Andrew be able to buy his first mobile phone?

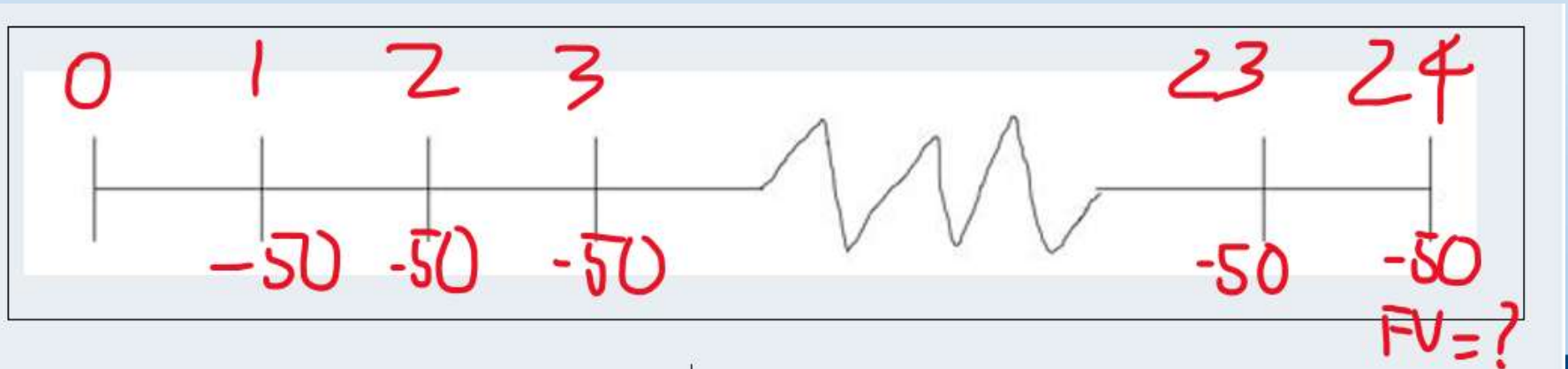
Step 2: Draw timeline

Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ? Would Andrew be able to buy his first mobile phone?

Step 2: Draw timeline



Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ? Would Andrew be able to buy his first mobile phone?

Step 3: Identify the appropriate formula to be applied here

Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ? Would Andrew be able to buy his first mobile phone?

Step 3: Identify the appropriate formula to be applied here

$$FV = PMT \left[\frac{(1+i)^n - 1}{i} \right]$$

Activity 3



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Step 4: Calculate and answer the question

Activity 3



Andrew is currently depositing \$50 each month for the next 2 years with the hope to buy his first mobile phone which is expected to cost \$1,300 in 2 years' time. If the current interest rate of 5% p.a is held for the whole time period, how much Andrew is able to save after the time period ? Would Andrew be able to buy his first mobile phone?

Step 4: Calculate and answer the question

$$FV = 50 \times \left[\frac{\left(1 + \frac{5\%}{12}\right)^{24} - 1}{\frac{5\%}{12}} \right] = 1259.30$$

Andrew won't be able to buy his phone after 24 months

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

Question 1: Please advise Emma on how much is her monthly repayment?

Step 1: Identify the known variables and unknown variables

$I =$

$N =$

$PMT =$

$PV =$

$FV =$

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

Question 1: Please advise Emma on how much is her monthly repayment?

Step 1: Identify the known variables and unknown variables (5 minutes)

$$I = 6\%/12$$

$$N = 30 \times 12$$

$$PMT = ?$$

$$PV = 80\% \times 800,000 = 640,000$$

$$FV = 0$$

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

Question 1: Please advise Emma on how much is her monthly repayment?

Step 2: Draw timeline

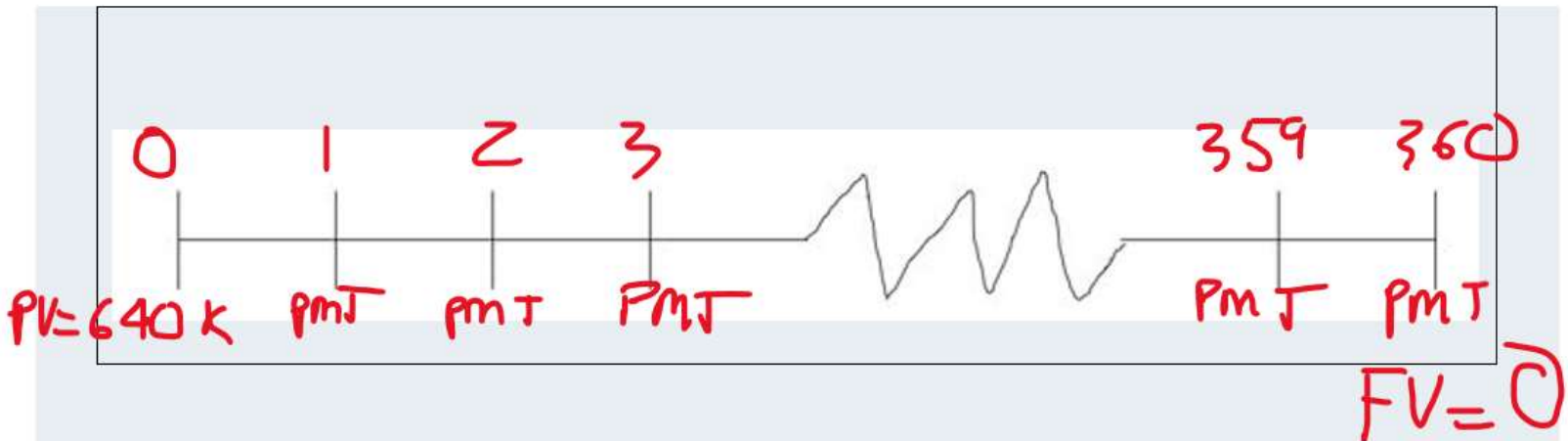
Activity 4



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Step 2: Draw timeline



Activity 4



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Question 1: Please advise Emma on how much is her monthly repayment?

Step 3: Identify the appropriate formula to be applied here

Activity 4



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Step 3: Identify the appropriate formula to be applied here

Since we know PV, we will use the PV annuity formula to calculate PMT:

From:

$$PV = PMT \times \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

To:

$$PMT = \frac{PV}{\left[\frac{1 - (1+i)^{-n}}{i} \right]}$$

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

Question 1: Please advise Emma on how much is her monthly repayment?

Step 4: Calculate and answer the question

Activity 4



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Question 1: Please advise Emma on how much is her monthly repayment?

Step 4: Calculate and answer the question

$$PMT = \frac{640,000}{\left[\frac{1 - (1 + 6\%/12)^{-360}}{6\%/12} \right]} = 3837.12$$

The monthly repayment is 3837.12 and after 30 years, Emma will pay off her home mortgage (assuming the interest rate is fixed for the whole 30 years- which is unlikely)

Activity 4



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Question 2: How much can she borrow if she could only afford to pay \$3000 per month (assuming the same 30-year loan period and interest rate of 6% compounding monthly)

Step 1: Identify the known variables and unknown variables

Activity 4



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Step 1: Identify the known variables and unknown variables

$$I = 6\%/12$$

$$N = 30 \times 12$$

$$PMT = 3000$$

$$PV = ?$$

$$FV = 0$$

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

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Step 2: Draw timeline

Activity 4



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Step 2: Draw timeline



Activity 4



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Step 3: Identify the appropriate formula to be applied here

Activity 4



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Question 2: How much can she borrow if she could only afford to pay \$3000 per month (assuming the same 30-year loan period and interest rate of 6% compounding monthly)

Step 3: Identify the appropriate formula to be applied here

$$PV = PMT \times \left[\frac{1 - (1 + i)^{-n}}{i} \right]$$

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

Question 2: How much can she borrow if she could only afford to pay \$3000 per month (assuming the same 30-year loan period and interest rate of 6% compounding monthly)

Step 4: Calculate and answer the question

Activity 4



Emma is looking to buy her first home to live in. She is currently looking at a house that is priced at \$800,000. She is planning to borrow 80% of the house price and wondering how much her monthly mortgage payment is, assuming the loan will be paid off completely after 30 years. Let's assume the current interest rate of 6% is held for the whole 30 years,

Question 2: How much can she borrow if she could only afford to pay \$3000 per month (assuming the same 30-year loan period and interest rate of 6% compounding monthly)

Step 4: Calculate and answer the question

$$PV = 3000 \times \left[\frac{1 - \left(1 + \frac{6\%}{12}\right)^{-360}}{\frac{6\%}{12}} \right] = 500,374.84$$

Emma can only borrow roughly 500,374.84 if she could only afford to pay 3,000 per month



Fundamentals of Business Finance

Tutorial B

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 1: Identify the variables:

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 1: Identify the variables:

$$N = 4 \times 2$$

$$I = 4\%/2$$

$$PV = ?$$

$$FV = 1,000$$

$$PMT = 50$$

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

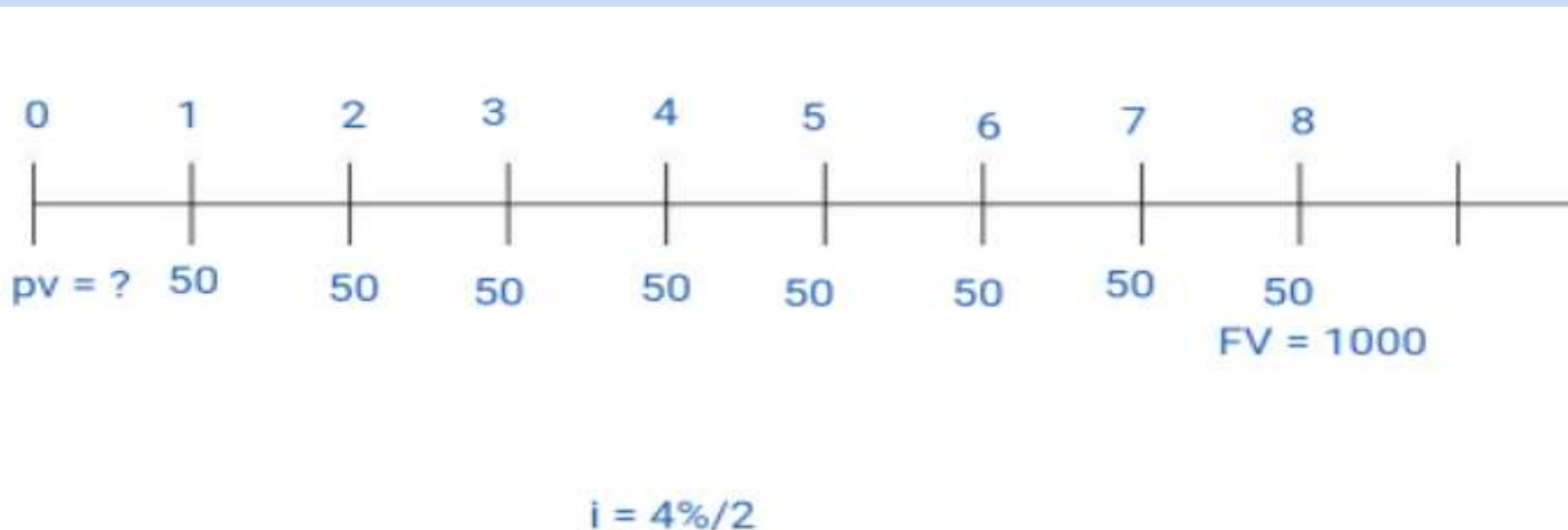
Step 2: Draw the timeline

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 2: Draw the timeline



Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 3: Identify the appropriate formula to be applied here

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 3: Identify the appropriate formula to be applied here

PV of annuity + PV of lump sum

$$PV = PMT \left[\frac{1 - (1 + i)^{-n}}{i} \right] + PV = \frac{FV}{(1 + i)^n}$$

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 4: Calculate the Present Value of the annuity

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 4: Calculate the Present Value of the annuity

$$PV = 50 \times \left[\frac{1 - \left(1 + \frac{4\%}{2}\right)^{-8}}{\frac{4\%}{2}} \right] = 366.27$$

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 5: Calculate the Present Value of the lump sum

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 5: Calculate the Present Value of the lump sum

$$PV = \frac{1000}{\left(1 + \frac{4\%}{2}\right)^8} = 853.49$$

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 6: Calculate the Present Value of the product

Activity 1



An investment product is going to provide you an income of \$50 every half year for the next 4 years. At the end of year 4, the investment will provide you a lump sum of \$1,000. If the interest rate for the whole time period is fixed at 4% pa, what is a fair price (Present Value) of the product?

Step 6: Calculate the Present Value of the product

$$PV = 366.27 + 853.49 = 1219.76$$